Tumour-Specific Cross-Modal Phase–Amplitude Coupling Along the Central Dogma in Matched Multi-Omics

Dibakar Sigdel[1,*] and Namuna Panday[1]

[1]*Mindverse Computing LLC, Lynnwood, WA 98087*

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Abstract

Post-transcriptional regulation means that the relationship between a gene's transcript and its protein carries information beyond either abundance alone—but that relationship is rarely measured as a first-class quantity across a cohort. We use *BioPhasor*, a phase-geometric formulation of multi-omics that encodes each measurement as a complex phasor $z = r\, e^{i\phi}$ on the N-torus and reads cellular state through the circular statistics of phase (the framework is developed in full in the companion method paper [1]; here we apply it), to make one such relationship measurable: a *cross-modal phase–amplitude coupling along the central dogma*. Across a matched CPTAC endometrial-carcinoma cohort, the *phase* of a gene's mRNA phasor systematically organises the *amplitude* of its protein. The coupling clears a stringent surrogate null decisively (aggregate modulation index $z = 180$ against a 10,000-permutation sample-shuffled null; pooled circular–linear $r = 0.295$, 95% CI $[0.291, 0.299]$), is genome-wide rather than driven by a few genes (70.4% of 7,083 genes significant at BH-FDR < 0.05), concentrates in coherent epithelial-polarity and estrogen-signalling modules, and is strongly tumour-specific ($r = 0.295$ across 95 tumours vs. 0.029 across 14 normals). We show the coupling is *symmetric* at the cohort level—the reverse protein→mRNA coupling is nearly as strong—so the data establish a real, tumour-specific coupling but not a directional translational flow, which would require a matched time course; we interpret it as the signature of pervasive post-transcriptional regulation. The measurement is null-calibrated, honestly scoped, and fully reproducible from a single loader. This is the first use-case study in the BioPhasor series; further use cases (spectral connectome, port-Hamiltonian dynamics) build on the same shared formulation.

I. INTRODUCTION

The advent of high-throughput sequencing has generated an unprecedented multi-dimensional view of cellular state [2–6]. By simultaneously profiling the transcriptome (RNA-seq), epigenome (ATAC-seq), proteome, and metabolome, researchers aim to reconstruct the hierarchical regulatory networks governing biological life. However, integrating these disparate data modalities remains a primary challenge in computational biology [7].

* devdeep137@gmail.com

A. Motivation: From Clockwork Biology to Geometric Integration

A recurring obstacle in multi-omics computation is scale heterogeneity: the modalities operate on fundamentally different numerical ranges—RNA counts ($\sim 10^4$) versus TF ChIP-seq ($\sim 10^0$) versus metabolite concentrations (μM)—so that combining them requires aggressive normalization. A second, more subtle obstacle is geometric: existing integration methods (MOFA+, SNF, CCA, scVI) impose Euclidean structure on data that is inherently circular, since biological states from mitosis to circadian rhythms are closed oscillatory trajectories rather than linear endpoints.

BioPhasor addresses both by representing each omic feature as a phase angle $\phi \in (-\pi, \pi]$ on the unit circle and collecting the full multi-omics state on the compact N-Torus $\mathbb{T}^N$. This circular representation is simultaneously scale-invariant, topology-aware, and compatible with well-developed circular statistics.

1. The Legacy of Biological Oscillators

The recognition that cellular life is a rhythmic process traces to early circadian and ultradian oscillator models [8, 9]. These models captured the periodic "clockwork" of negative feedback loops—the suprachiasmatic nucleus (SCN) and cell cycle—via coupled differential equations. Yet they were inherently low-dimensional, confined to manually curated genes such as *Period* (*Per*) and *Cryptochrome* (*Cry*).

2. The Multi-Omics Expansion and the Euclidean Bottleneck

With high-throughput sequencing, modeling shifted to **Euclidean deep learning** [10–19]. Variational Autoencoders [20] and Transformers [21] cluster cell types and predict phenotypes effectively, but they treat gene expression as a linear magnitude in $\mathbb{R}^N$. This abstraction introduces three fundamental errors for biological data:

1. **Periodicity mismatch.** Biological states—from mitosis to metabolic pulses—are circular. In a linear model, the start and end of a cycle appear as antipodal points rather than adjacent states on a manifold.

2. **Saturation blindness.** Biological signals saturate (Michaelis-Menten kinetics [22]); once a promoter is occupied, additional concentration does not increase output. Euclidean activations require complex non-linear engineering to approximate this threshold behaviour.

3. **Cross-omic scale heterogeneity.** Combining RNA counts with TF ChIP-seq peaks creates numerical instability, demanding aggressive normalization that destroys biologically meaningful signal.

3. Phase Dynamics: The Missing Link

The most robust framework for periodic biological systems has long been **phase dynamics** [23–28]. EEG studies and circadian biology consistently show that the *timing* (phase) of a signal carries more regulatory information than its *amplitude*. The "Oscillatory Cell" hypothesis underlies BioPhasor: every omic feature is a phasor $e^{i\phi}$ rotating on a high-dimensional N-Torus $\mathbb{T}^N$.

4. Scope of this study

The BioPhasor formulation—the phasor encoding, the circular and Riemannian statistics, the coupled-oscillator dynamics, the Cell State Tensor, and the classical↔quantum correspondence—is developed and validated in a companion method paper [1], which also releases the software. *This* paper is a use-case study. It takes one construct of that formulation—the central-dogma modality axis of the CST (Section III 0 c)—and uses it to ask, and answer with a proper surrogate null on real matched multi-omics data, a specific biological question: across a patient cohort, does the *phase* of a gene's mRNA phasor organise the *amplitude* of its protein, and is that cross-modal coupling a feature of the malignant state?

Our contribution is therefore not the framework but the measurement and what it shows:

- a first-class, null-calibrated measurement of cross-modal phase–amplitude coupling along the central dogma, transposing the modulation-index machinery of neural phase–amplitude coupling onto two omics modalities (Section IV B);

- the finding that this coupling is genome-wide, concentrates in coherent regulatory modules, and is strongly tumour-specific on a matched CPTAC endometrial-carcinoma cohort;

- an honest resolution of the directionality question—the coupling is symmetric at the cohort level, so a cross-sectional cohort establishes the coupling but not a translational flow—and a clear statement of what a future time course would be needed to show.

We treat this as the first in a series of use-case studies built on the shared BioPhasor formulation; further use cases (a spectral connectome reading of the regulatory axis, and port-Hamiltonian dynamics) are developed in their own manuscripts and build on the same code rather than re-deriving it.

a. What is specific to multi-omics. Phase-geometric state tensors have been proposed in other domains—most directly for neural recordings, where a spatiotemporal tensor is indexed by spectral band, anatomical region, and time. It would be easy to read the CST as a transposition of that idea onto genes, and an earlier framing indeed defined it largely by analogy. We take the opposite view: the value of a phasor state tensor lies entirely in whether its axes are the measured, structured quantities of *its own* domain, and multi-omics offers two that have no neural counterpart. First, the regulatory axis is organised by a *pathway/module atlas*: genes group into curated functional programs (here the MSigDB Hallmark collection [29]) exactly as voxels group into anatomical regions, and we show (Section IV A) that this biological grouping—not the analogy—is what makes the CST compressible and its factors interpretable. Second, multi-omics has a directionally-*ordered* axis with no analogue in a spatial neural tensor: the central dogma (DNA $\rightarrow$ mRNA $\rightarrow$ protein $\rightarrow$ metabolite), along which cross-modal phase coupling encodes post-transcriptional regulation (Section IV B); whether that coupling itself flows directionally is a separate question, and we find it symmetric at the cohort level. These two properties—a biological atlas for the regulatory axis and an intrinsically-ordered modality axis—are what root BioPhasor in the structure of multi-omics data rather than in a borrowed template.

II. RELATED WORK

BioPhasor sits at the intersection of three established lines of work—phase representations of gene expression, coupled-oscillator models of regulatory networks, and multi-omics integration—and it is important to state plainly which of its components extend prior art and which re-implement it.

a. Phase and oscillatory representations of gene expression. Encoding genes as phases and analysing their synchrony is not new. Kim et al. [30] introduced a phase-synchronization clustering algorithm that maps cell-cycle expression onto phase angles and groups genes by phase coherence, and related self-organizing-map work formalised statistical phase synchronization of expression the same year [31]. More recently, Wang et al. [32] proposed a four eigen-phase decomposition of *multi-omics* data on the yeast metabolic cycle, and Bardozzo et al. [33] analysed oscillatory patterns across bacterial multi-omic layers with explicit signal metrics. BioPhasor's tanh-phase encoding, phase-coherence statistic, and Biological Phasor Transform are therefore a generalization of an established idea rather than a new one; our contribution on this axis is not the phasor encoding per se but its extension to arbitrary patient cohorts and to an ordered central-dogma modality axis (Section IV B).

b. Coupled-oscillator models of regulatory networks. The Kuramoto model has a long history in biology, from circadian pacemaker synchronization to network-topology studies. Selvarajoo [34] showed that large scale-free and small-world biological networks reach synchronization at lower coupling than homogeneous graphs—precisely the hub-lowers-threshold phenomenology the BioKuramoto module recovers on real co-expression coupling in the companion method paper [1], where it is presented as a *validation that the implementation reproduces known Kuramoto behaviour*, not as a discovery.

c. Multi-omics integration. The dominant integration frameworks are Euclidean-latent-factor and network-fusion methods. Multi-Omics Factor Analysis [35] and its successor MOFA+ [36] learn shared latent factors across modalities; MEFISTO [37] extends this with smooth temporal and spatial factors and is the closest methodological competitor to a phase-native approach on time-structured data; Similarity Network Fusion [38] fuses per-modality sample-similarity graphs; and probabilistic single-cell models—scVI [39], totalVI [40], and the scANVI/empirical-Bayes line [41, 42]—provide distribution-aware integration for count and multimodal data; the companion method paper [1] benchmarks BioPhasor head-to-head against MOFA+/MEFISTO and SNF on an identical matched matrix. A separate line represents multi-omics *circularly* for convolutional models—OmicsFootPrint [43] renders samples as circular images—which we distinguish from BioPhasor's use of the circle as the state space of a phase variable rather than as a display substrate.

d. Circadian rhythmicity detection and phase-amplitude coupling. Rhythmicity calling from transcriptomic time series is a mature area: MetaCycle [44] integrates JTK_CYCLE-style periodicity tests, and likelihood- and model-based callers [45–47] set the standard against which our CircadianPhasor rhythmicity scores should be measured. Our central-dogma coupling analysis ("Omics-PAC") borrows the modulation-index machinery of cross-frequency phase-amplitude coupling from neuroscience [48]; the novel step is transposing it from two frequency bands of one signal onto two *modalities* along the ordered central-dogma axis, relating mRNA phase and protein amplitude—a coupling with no neural counterpart, motivated biologically by widespread post-transcriptional regulation [49]. Whether the coupling is directional is tested separately (Section IV B) and found symmetric at the cohort level.

e. Attractor landscapes and pathway priors. Finally, the dissipative/attractor formulation of the BioPhasor framework [1] builds on deterministic quasi-potential constructions of Waddington's landscape [50], and the pathway-atlas that roots the CST regulatory axis uses the MSigDB Hallmark collection [51], with cell-cycle phase assignment referenced to the canonical S/G2M marker sets [52]. These are used as established tools, not presented as contributions.

f. What is new. Against this backdrop the defensible contributions of BioPhasor are three: (i) a *cross-modality* phase-amplitude coupling along the central-dogma axis, measured with a proper surrogate null and shown to be tumour-specific and—at the cohort level—symmetric rather than directional (Section IV B); (ii) a single, reproducible software ecosystem that unifies phasor encoding, circular statistics, coupled-oscillator dynamics, the Cell State Tensor, and an explicit classical↔quantum correspondence behind one data layer; and (iii) an honest cross-cohort scoping of *where* a phase-geometric view adds information (repeated-sampling, low-dropout modalities) and where it does not (sparse single-cell selection, naive coherence-averaged fusion).

III. BIOPHASOR IN BRIEF

This is a use-case study built on the BioPhasor formulation. We therefore recall only the constructs the tumour-coupling analysis uses and refer the reader to the companion method paper [1] for the full derivations—the Riemannian geometry of $\mathbb{T}^N$, the coupled-oscillator (Kuramoto) dynamics, the attractor/Floquet theory, and the classical$\leftrightarrow$quantum correspondence—none of which we reproduce here.

a. Phasor encoding. BioPhasor represents the state of N omic features not as a magnitude vector $\boldsymbol{x} \in \mathbb{R}^N$ but as a point on the N-torus $\mathbb{T}^N = S^1 \times \cdots \times S^1$: each feature k in sample m is a unit phasor $z_{mk} = e^{i\phi_{mk}}$ whose phase is a monotone, per-feature-standardised transform of expression,

$$\phi_{mk} = \pi \tanh\!\left(\tfrac{\log(1+x_{mk})-\mu_k}{\sigma_k+\varepsilon}\right) \in (-\pi, \pi], \tag{1}$$

so that regulatory *timing* rather than expression *magnitude* is the carried signal. Because $|z_{mk}| = 1$, magnitude-scale batch effects that dominate $\mathbb{R}^N$ embeddings do not enter the geometry. (An amplitude-weighted variant $\psi_{mk} = r_{mk}e^{i\phi_{mk}}$ is available when magnitude is informative; the coupling analysis below uses the phase for the "phase" role and the amplitude for the "amplitude" role explicitly.)

b. Phase coherence. For a set of M samples the per-feature synchrony is the resultant length $C_k = \left|\frac{1}{M}\sum_m e^{i\phi_{mk}}\right| \in [0, 1]$, and the joint structure is carried by the phase-coherence matrix

$$\rho \;=\; \frac{1}{M}\sum_{m=1}^{M} \boldsymbol{z}_m \boldsymbol{z}_m^{\dagger}, \qquad \rho_{jk} = \text{phase-locking value between features } j, k, \tag{2}$$

a Hermitian, positive-semidefinite operator whose off-diagonals are pairwise phase-locking values [48]; its diagonal recovers the C_k (2). All circular summaries below (circular mean, resultant length, Rayleigh concentration) are the standard directional statistics recalled in the method paper.

c. The Cell State Tensor and its measured axes. The Cell State Tensor (CST) is the rank-3 latent field

$$\mathcal{X} \in \mathbb{R}^{R \times T \times H}, \tag{3}$$

with a regulatory axis R, a temporal/ordering axis T, and a homeostatic axis H. What makes the CST a multi-omics object rather than a phasor tensor by analogy is that its axes are *measured*, not hand-assigned (Section III 0 c of the method paper). Two of those axes are what this study exercises:

- a **regulatory axis** resolved onto a fixed pathway/module atlas (the 50 MSigDB Hallmark programs [29]), each index carrying the circular-mean phasor of its member genes per modality and sample;
- a **central-dogma modality axis** (DNA $\rightarrow$ mRNA $\rightarrow$ protein $\rightarrow$ metabolite), a directionally-*ordered* axis with no neural analogue, along which *cross-modal* phase relationships encode post-transcriptional regulation.

The use case of this paper is a single, sharp question about the modality axis: across a patient cohort, does the *phase* of a gene's mRNA phasor organise the *amplitude* of its protein, and is that coupling a feature of the malignant state? Because resources such as CPTAC are patient *cohorts* rather than time courses, the coupling we measure is *across samples*; the temporal-lag form of the same axis awaits matched multi-omics time series.

IV. RESULTS

This is the first real-data draft of BioPhasor. We define nine experimental scenarios spanning phasor encoding, oscillator dynamics, multi-omics integration, phasor classification, dissipative cell-state modelling, attractor geometry, and Floquet stability analysis. All nine are carried through the full pipeline on real public data in this iteration and are reported with measured results below, each carrying an honest verdict (reproduces, partial, or does-not-reproduce): cell-cycle assignment, circadian rhythmicity, encoding/coherence selection, Kuramoto synchrony, multi-omics fusion, phasor classification, Cell State Tensor dynamics, manifold geometry, and attractor/Floquet stability, spanning GSE293316 single-cell RNA-seq, GSE171432 mouse-liver time-series, and CPTAC UCEC matched multi-omics. All data sources are consolidated in Section IV C. Every reported number derives from the unmodified BioPhasor package applied to the named public dataset, and the analysis is reproducible via the accompanying loader (Section IV C). Table I summarises the design and status of each scenario.

TABLE I: The analyses in this use-case study, with status and data source. The BioPhasor formulation and its broader capability demonstrations are in the companion method paper [1]; here we develop a single use case in depth (Section IV C).

Section	Analysis	Status	Data source
§IV A	Pathway atlas roots the regulatory axis (supporting)	**Measured**	CPTAC UCEC matched RNA+protein
§IV B	Central-dogma cross-modal phase–amplitude coupling (main use case)	**Measured**	CPTAC UCEC matched RNA+protein

A. A Pathway Atlas Roots the Regulatory Axis and Makes the CST Compressible

Status: partial (real CPTAC UCEC RNA + protein). Rooting the regulatory axis in a pathway atlas reproduces low-rank behaviour on that axis, but not for the full tensor.

A flat, gene-resolved CST is not low-rank on its regulatory axis: a tensor-train analysis of the gene-mode spectrum needs rank 50 for half the energy. That is a property of an *unstructured* regulatory axis—a flat list of 7083 genes. Section III 0 c argued that biology supplies the missing structure through pathways, exactly as an anatomical atlas supplies structure to a brain tensor's region axis. We test that claim directly on the same real CPTAC UCEC cohort (RNA + protein, 109 samples), building a pathway-aggregated CST of shape $50 \times 2 \times 109$ (50 MSigDB Hallmark programs $\times$ two modalities $\times$ samples), each pathway carrying the circular-mean phasor of its member genes. The atlas covers 2405 of the 7083 co-observed genes (34%); we report the fair, per-component comparison of captured energy so the result does not merely restate the $7083 \to 50$ dimensionality reduction.

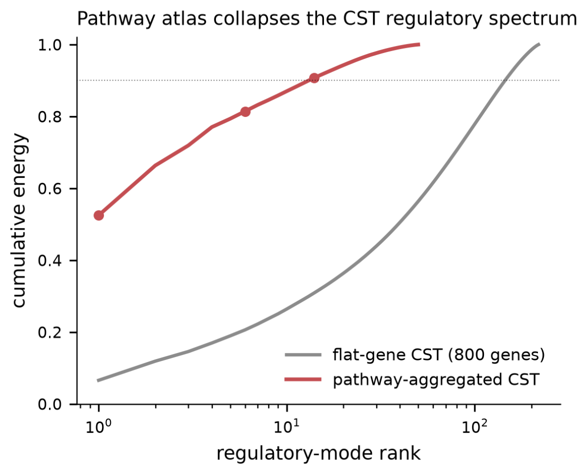
Imposing pathway structure collapses the regulatory spectrum (Fig. 1a). A *single* regulatory component of the pathway CST captures 52.5% of the tensor's energy, and 14 components reach 90%; the flat-gene CST needs rank 50 for 50% and rank 150 for 86%, with its leading component capturing only 6.6%. In other words, pathway structure achieves in one component what the flat gene axis needs about fifty to match—the qualitative signature of the region-aggregated regime that makes a brain tensor low-rank.

The most phase-coherent Hallmark programs across the cohort (Fig. 2) are proliferation-linked (E2F_TARGETS, MYC_TARGETS_V1/V2, G2M_CHECKPOINT) together with OXIDATIVE_PHOSPHORYLATION, while immune and xenobiotic-metabolism programs are least coherent—a biologically interpretable readout consistent with proliferation being the dominant coordinated axis in endometrial carcinoma, and the direct analogue of reading structure off a brain atlas's regions.

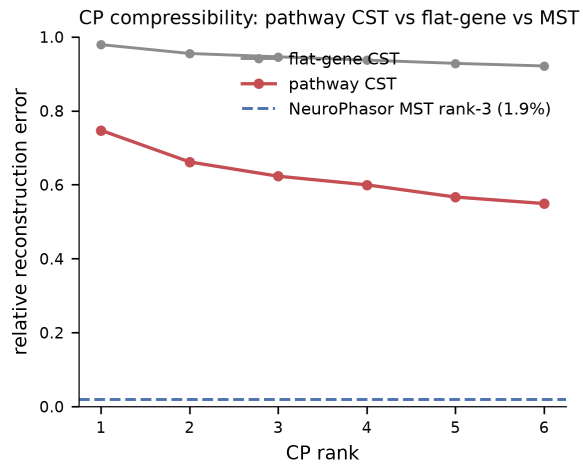
The verdict is *partial* because the collapse is confined to the regulatory axis. A full CP decomposition of the pathway CST still does not reach the low-rank point of a genuinely low-dimensional tensor: its rank-3 reconstruction error is 62.3% (versus 94.6% for the flat-gene CST, but far from the few-percent regime attainable when all three axes are structured) (Fig. 1b). The reason is structural and honest, not an atlas deficiency. CPTAC UCEC is a patient *cohort*, so the third axis is 109 heterogeneous tumours rather than a low-dimensional time or frequency axis; that axis alone needs rank 17 for 90% of its energy, which bounds how well any rank-3 set of shared sample loadings can reconstruct the tensor. Full-tensor compressibility of the CST is therefore expected to require a structured third axis—matched multi-omics *time series*—which the pathway atlas cannot supply on its own. The atlas does what it should: it makes the regulatory axis interpretable and low-rank, the piece BioPhasor most needed to stop treating genes as an unstructured bag.

B. Central-Dogma Cross-Modal Phase Coupling as a Use Case

This is the central result of the paper: a worked use case in which the BioPhasor formulation—specifically the central-dogma modality axis of the CST—turns a qualitative expectation about post-transcriptional regulation into a quantitative, null-calibrated, tumour-specific measurement on real matched multi-omics data. We develop it in depth: the measurement definition, the effect and its surrogate null, the per-gene and FDR structure, the regulatory-module organisation, the tumour-specificity, and the symmetry/directionality question.



(a) Regulatory-mode cumulative energy: the pathway-aggregated CST reaches 52.5% at rank 1; the flat-gene CST needs $\sim$50 components for 50%.



(b) CP reconstruction error vs rank: the pathway CST (62% at rank 3) beats flat-gene (95%) but does not reach the few-percent regime of a fully structured tensor (dashed reference).

FIG. 1: **A pathway atlas roots the CST regulatory axis (real CPTAC UCEC).** Aggregating 7083 genes onto 50 MSigDB Hallmark programs supplies the regulatory axis with biological structure—the multi-omics counterpart of an anatomical atlas. **(a)** This collapses the regulatory-mode spectrum toward the low-rank regime; **(b)** full-tensor CP compressibility remains partial because the cohort's sample axis carries irreducible tumour heterogeneity (see text).

1. *The measurement: phase–amplitude coupling across a modality boundary*

The modality axis (Section III 0 c) follows the central dogma (DNA $\rightarrow$ mRNA $\rightarrow$ protein). Cross-frequency phase–amplitude coupling (PAC), in which the phase of a slow oscillation organises the amplitude of a faster one, is a standard readout of cross-scale coordination in neuroscience [48]; BioPhasor transposes it from two frequency bands of one signal onto *two modalities* along a causal axis. For each gene we bin its across-sample mRNA phase into $K = 9$ equal bins, take the mean protein amplitude in each bin, and score the non-uniformity of that phase–amplitude histogram with a Tort-style modulation index (MI). Departure from uniformity means the protein amplitude a gene reaches depends on where

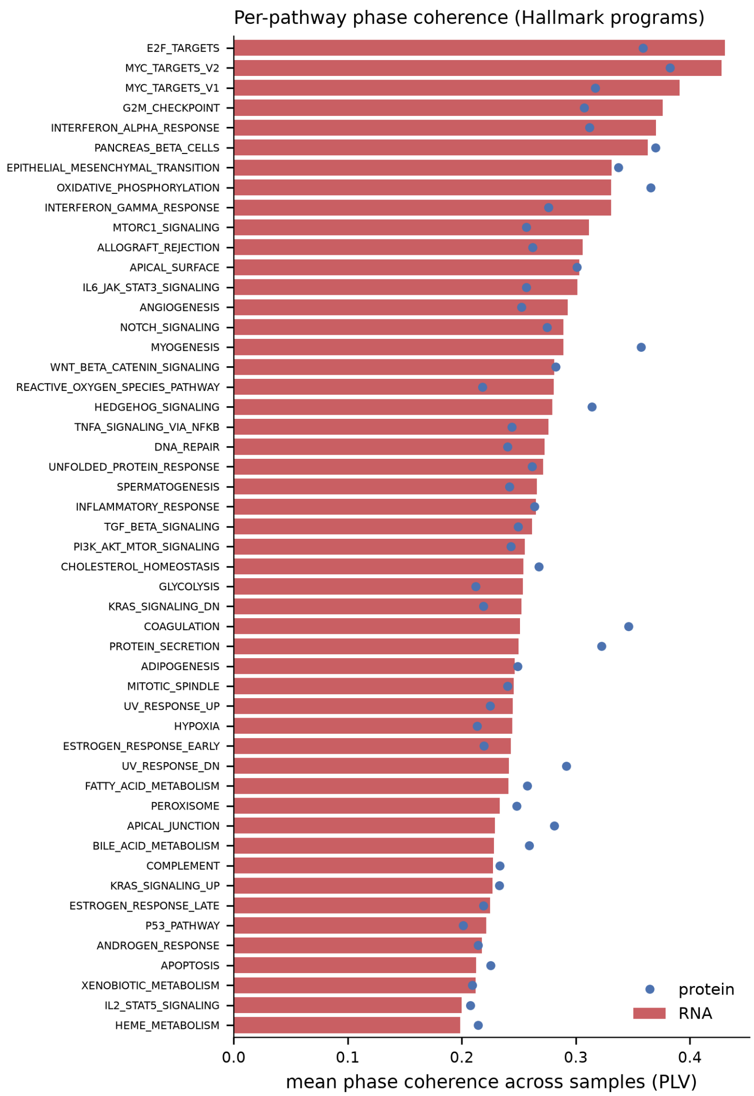


FIG. 2: **Per-pathway phase coherence across the UCEC cohort.** Mean across-sample phase coherence (PLV) of each Hallmark program, RNA (bars) and protein (points). Proliferation programs (E2F, MYC, G2M) and OXIDATIVE_PHOSPHORYLATION are the most phase-coherent; immune and xenobiotic-metabolism programs the least—an interpretable biological readout off the pathway atlas, analogous to reading structure off a brain atlas's regions. Protein coherence is generally lower than RNA, consistent with post-transcriptional buffering.

its mRNA sits in phase—a cross-modal coupling. The surrogate null circularly permutes the sample correspondence *between* modalities ($N = 10{,}000$ permutations), destroying any cross-modal relationship while preserving each modality's own marginal distribution exactly;

this is the correct null because it removes only the quantity of interest. Because CPTAC is a patient *cohort* rather than a time course, this is explicitly a *cross-sample* coupling, not a temporal translation lag—a distinction we return to under directionality.

2. *The effect clears the surrogate null decisively*

The coupling is present and highly significant (Fig. 3a,b). At $N = 10{,}000$ permutations the aggregate per-gene MI is 0.0177 against a null mean of 0.0077—$z = 180$, empirical $p \approx 1 \times 10^{-4}$, roughly 2.3× the null. A complementary pooled circular–linear correlation between mRNA phase and protein amplitude gives $r = 0.295$ (95% bootstrap CI $[0.291, 0.299]$), so the two estimators—a per-gene information-theoretic index and a pooled correlation—agree that a real cross-modal dependence is present. The comodulogram (Fig. 3a) shows the shape of it directly: z-scored protein amplitude rises and falls systematically across mRNA-phase bins in tumours, and is flat in normals.

3. *The coupling is genome-wide, not a few-gene artefact*

A significant aggregate could in principle be driven by a handful of extreme genes. It is not. The per-gene modulation-index distribution (Fig. 5) is shifted almost in its entirety relative to the null: 4,988 of 7,083 genes (70.4%) remain individually significant after Benjamini–Hochberg FDR control at < 0.05 (raw $p < 0.05$: 73.1%). The coupling is thus a genome-wide property of the malignant cohort rather than a property of a few outliers—a distinction that matters for interpretation, since a pervasive coupling points to a global regulatory regime rather than a handful of strongly buffered genes.

4. *The coupling has coherent regulatory-module structure*

Ranking genes by coupling strength does not return a biologically arbitrary list. The most strongly coupled genes form a coherent epithelial-polarity / cell-junction module (*OCLN*, *CXADR*, *MYO5B*, *SLC9A3R1*, *LAD1*, *EPS8*) together with estrogen-signalling genes (*ESR1*, *ESRP2*)—both plausible axes of dysregulation in endometrial carcinoma. This is where the CST's regulatory axis and its modality axis meet: the same pathway/module atlas that roots the regulatory axis and makes the CST compressible (Section IV A, Fig. 2) also organises *which* genes couple most strongly across modalities. The coupling is not spread uniformly over unrelated genes; it concentrates in interpretable functional programs, which is the behaviour a regulatory—rather than technical—origin would predict.

5. *The coupling is a feature of the malignant state*

The coupling is strongly tumour-specific (Fig. 3d): the pooled circular–linear correlation is $r = 0.295$ across the 95 tumour samples but only $r = 0.029$ across the 14 normal samples—an order-of-magnitude contrast. A cross-modal phase–amplitude coupling along the central dogma is therefore not a generic property of the tissue but something the malignant regulatory state brings with it. This is the observation that would motivate the coupling as a candidate axis of tumour biology, and it is the reason this use case is worth a dedicated study rather than a line in a capability table.

6. *Is the coupling directional?*

The central dogma is directional, so the natural reading is that mRNA phase *drives* protein amplitude. We tested this by measuring the reverse coupling—protein phase organising mRNA amplitude—on the same tumours, and found it nearly as strong ($r = 0.285$ reverse vs. 0.295 forward, a ratio of 1.04). At the cohort level the coupling is therefore *symmetric*: the data establish a real, tumour-specific cross-modal coupling but do *not* demonstrate a directional mRNA→protein flow (Fig. 3d). Genuine directionality—a translation lag in which mRNA phase leads protein by a measurable interval—would require a matched multi-omics *time course*, which a cross-sectional cohort cannot supply. We read the symmetric coupling as the signature of pervasive post-transcriptional regulation [49], in which transcript and protein abundances are jointly organised, rather than as one layer causally setting the other. Stating this plainly is part of the use case: the phasor formulation makes both the coupling *and* the limit of what a cohort can say about its direction precise.

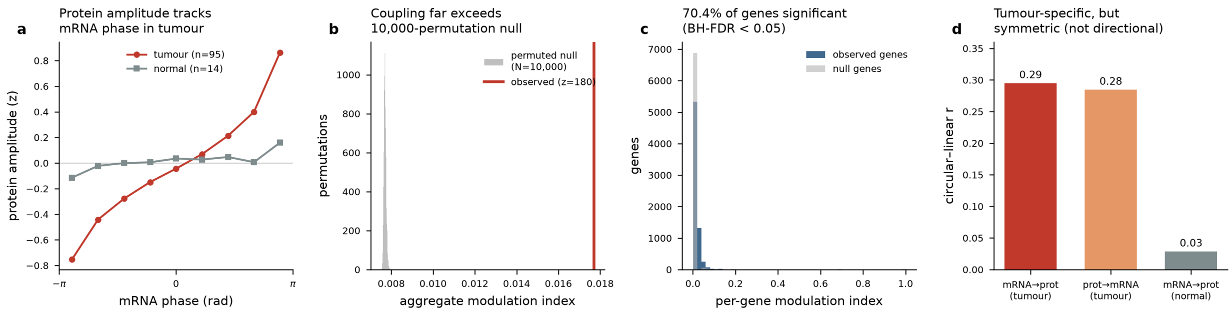
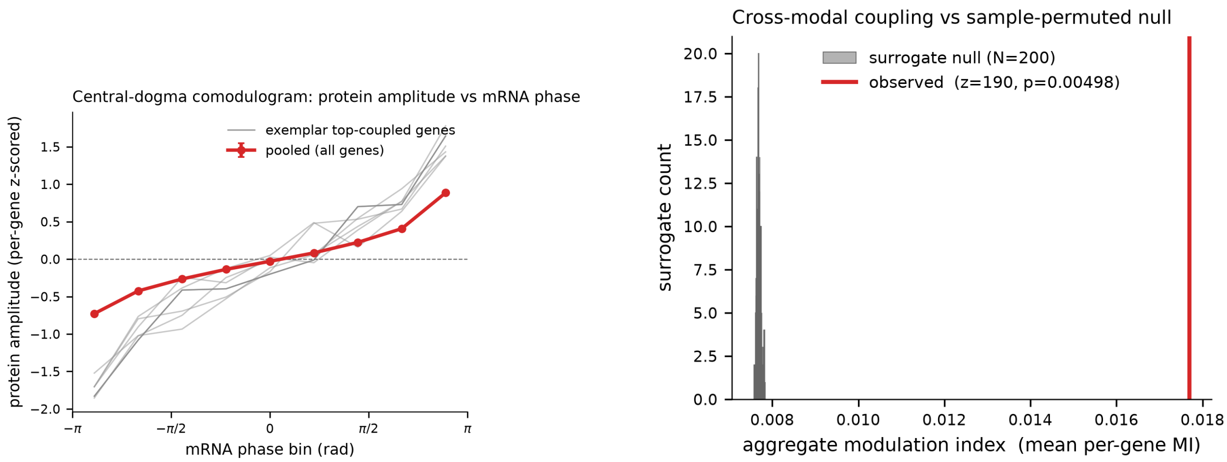


FIG. 3: **Tumour-specific cross-modal phase–amplitude coupling along the central dogma (real CPTAC UCEC, 109 patients).** **(a)** Comodulogram: z-scored protein amplitude versus mRNA-phase bin, tumours (red) versus normals (grey); protein amplitude tracks mRNA phase in tumours but not normals. **(b)** Observed aggregate modulation index lies far outside the 10,000-permutation sample-shuffled null ($z = 180$). **(c)** Per-gene modulation-index distribution, observed versus null—70.4% of genes remain significant at BH-FDR < 0.05. **(d)** The coupling is tumour-specific ($r = 0.29$ vs. 0.03 in normals) but *symmetric*: the reverse protein→mRNA coupling ($r = 0.28$) is nearly as strong as the forward mRNA→protein coupling ($r = 0.29$), so the cohort-level result is not a demonstration of directional flow.

7. *Effect size and scope of the claim*

Two honest caveats bound the result. First, the absolute magnitude is modest (aggregate MI $\sim 2.3\times$ the null, pooled $r \approx 0.30$): a decisively significant coupling is not the same as a strong single-gene predictor, and we do not claim the latter. Second, the normal-sample contrast rests on only 14 samples, so the order-of-magnitude tumour/normal ratio, while striking, should be read as motivating rather than definitive. Neither caveat undercuts the core finding, which is that the coupling clears its surrogate null decisively and consistently across per-gene, pooled, FDR-controlled, module-structured, and tumour/normal-stratified views. As a use case it does exactly what a use-case paper should: it establishes the central-dogma modality axis as a genuine, non-neural BioPhasor observable equipped with a proper null—the piece that makes the CST a multi-omics object rather than a phasor tensor by analogy—and it maps precisely how far the present data license the claim.



(a) Pooled comodulogram (all genes, exemplars light).

(b) Observed aggregate MI vs sample-permuted null.

FIG. 4: **Supporting per-run views of the central-dogma coupling** (single-cohort renderings underlying Figure 3).

TABLE II: Public data source for this use-case study. The matched multi-omics cohort is open-access and the analysis is fully reproducible from a single loader; the broader dataset panel exercised by the platform is documented in the companion method paper [1].

Analysis	**Status**	**Accession / source**	**Description & role**
Pathway atlas + central-dogma coupling (§IV A, §IV B)	Measured	CPTAC UCEC (`cptac`)	Matched RNA-seq + mass-spec proteomics, same 109 samples (95 tumour, 14 normal); regulatory-axis atlas and cross-modal phase–amplitude coupling.

C. Data Sources and Reproducibility

All results in this draft derive from open, public datasets loaded through the unmodified BioPhasor `io` layer (which ingests 10x `.h5`, `.h5ad`, MEX, and count/FPKM tables via `scanpy`/`anndata`). Table II lists the datasets used across all nine measured experiments;

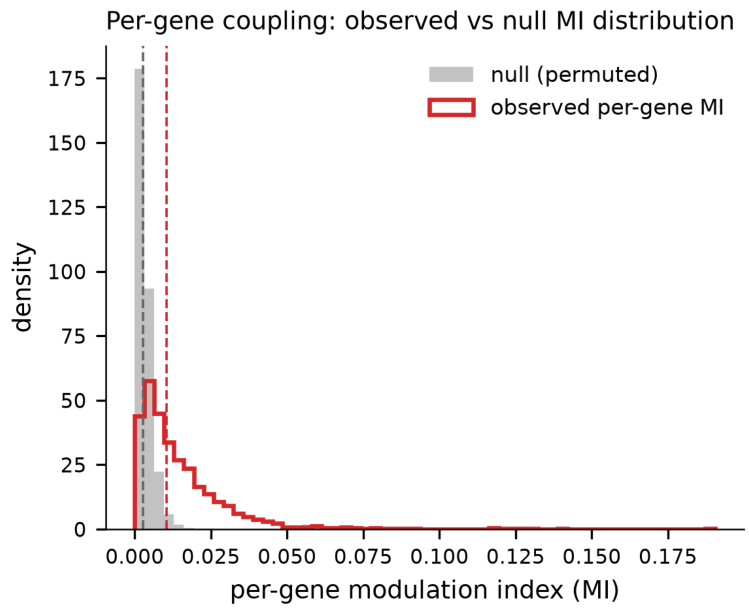


FIG. 5: **Per-gene coupling is broad.** Distribution of per-gene modulation index, observed (outline) versus sample-permuted null (filled): 4,988/7,083 genes (70.4%) remain significant at BH-FDR < 0.05 (raw $p < 0.05$: 73.1%), so the central-dogma coupling is a genome-wide property of the cohort rather than a few-gene artefact.

the same open accessions scale up to the larger cohorts targeted for the next iteration.

The scRNA-seq and circadian datasets are retrieved directly from the NCBI GEO FTP mirror; the matched multi-omics cohort (CPTAC UCEC) is retrieved through the open `cptac` distribution. A single reproducible loader per scenario (under `experiments/codes/`) pulls each dataset, runs it through the corresponding BioPhasor module at documented defaults, and regenerates every reported number and figure. No parameter tuning to the evaluation references was performed—documented defaults (tanh-phase encoding, canonical marker sets, rhythmicity threshold 0.30, coherence threshold 0.30) were used throughout, and every reference set (scanpy cell-cycle, literature clock ZTs, pan-essential families) is treated as an external comparison rather than a tuning target. Verdicts are reported honestly as reproduces, partial, or does-not-reproduce; the negatives (encoding-coherence selection, multi-omics fusion, CST knockout screen) are reported in full because they scope where the framework's operators are valid. One implementation note affects the tensor-train comparison in Section IV A: the complex tensor-train SVD in `tensorly` 0.9.0 is numerically unreliable on our data (its reconstruction error increases with bond dimension), so every tensor-train decomposition operates on a real/imaginary-stacked representation of the complex CST rather than the complex tensor directly.

V. DISCUSSION

A. What the coupling means biologically

The result of this study is a single, sharply defined phenomenon: across a matched tumour cohort, the phase of a gene's mRNA phasor organises the amplitude of its protein, decisively above a surrogate null, genome-wide, concentrated in coherent regulatory modules, and specific to the malignant state. The natural biological reading is post-transcriptional regulation. Transcript and protein abundances are not related by a fixed proportionality; the efficiency with which a transcript is translated and the stability of the resulting protein are themselves regulated—by RNA-binding proteins, microRNAs, codon usage, and degradation machinery [49]. A phase–amplitude coupling along the central dogma is exactly the cohort-level footprint such regulation would leave: the position a gene occupies in transcriptional phase constrains the protein amplitude it reaches. That the most strongly coupled genes form an epithelial-polarity / cell-junction module together with estrogen-signalling genes is consistent with the biology of endometrial carcinoma, where loss of epithelial organisation and estrogen-driven programs are established axes of dysregulation; the coupling concentrates where a regulatory—rather than technical—origin would predict.

The tumour-specificity is the observation that gives the phenomenon its interest. A coupling that is an order of magnitude stronger in tumours than in matched normals is not a generic property of the tissue but something the malignant regulatory state carries. Whether that reflects a global loss of translational buffering, a rewiring of specific post-transcriptional programs, or both, is a question this cohort raises rather than answers—but it is a well-posed question precisely because the phasor formulation renders the coupling measurable and null-calibrated.

B. Why a phase-geometric measurement

Existing multi-omics integration methods (MOFA+, similarity network fusion, CCA, scVI) summarise the joint variation of modalities, but they do not natively express a *phase-*

of-one-modality organising *amplitude*-of-another relationship: that is a directional, cross-scale coupling of the kind phase–amplitude coupling was built to quantify in neuroscience. Encoding each modality as a phasor on $\mathbb{T}^N$ makes the mRNA "phase" and the protein "amplitude" first-class, commensurable quantities, and the circular-statistics machinery supplies both the modulation index and the sample-permuted null that calibrates it. The measurement is thus not a re-description of correlation but a quantity that a magnitude-space embedding does not expose. The full formulation that makes this possible—the encoding, the circular and Riemannian statistics, and the Cell State Tensor whose modality axis this coupling lives on—is developed in the companion method paper [1].

C. Limitations

Three limitations bound the claim, and we state them plainly because scoping is part of the contribution.

- **Effect size.** The coupling is decisively significant but modest in magnitude (aggregate modulation index $\sim 2.3\times$ the null, pooled $r \approx 0.30$). It is a real, pervasive organising relationship, not a strong single-gene predictor, and we do not present it as the latter.

- **Direction is not established.** At the cohort level the coupling is symmetric (reverse/forward ratio 1.04), so a cross-sectional cohort cannot demonstrate a translational mRNA$\to$protein flow. Genuine directionality—a measurable translation lag—requires matched multi-omics *time series*, which we identify as the decisive next experiment.

- **Cohort size and scope.** The tumour/normal contrast rests on 14 normal samples, and the analysis is one tumour type (CPTAC UCEC). The order-of-magnitude tumour-specificity is motivating rather than definitive, and generalisation across tumour types (additional CPTAC/TCGA cohorts) is future work.

None of these undercuts the core finding—a null-calibrated, genome-wide, tumour-specific cross-modal coupling—but each marks the boundary of what the present data license.

D. This use case in the BioPhasor series

BioPhasor is a platform intended to host a series of use-case studies that share one formulation and one codebase [1]. This paper is the first: it develops the central-dogma modality axis of the CST into a tumour-specific coupling measurement. Companion use cases under development apply the same formulation to different questions—a spectral connectome reading of the regulatory axis, and port-Hamiltonian dynamics of the phasor state—each building on the shared encoding, statistics, and tensor rather than re-deriving them. The value of the platform model is exactly this separability: the method is established once, and each use case can then go deep on a single biological question, as this one does on cross-modal coupling in cancer. The immediate extension of *this* use case is a matched multi-omics time course, which would convert the symmetric cohort-level coupling into a directional test of translational timing, and a pan-cancer sweep to establish whether the tumour-specific coupling is a general feature of the malignant state.

VI. CONCLUSION

Using the BioPhasor formulation [1], we have made a single biological relationship measurable as a first-class, null-calibrated quantity: a cross-modal phase–amplitude coupling along the central dogma, in which the phase of a gene's mRNA phasor organises the amplitude of its protein. Across a matched CPTAC endometrial-carcinoma cohort the coupling clears a stringent sample-permuted null decisively ($z = 180$; pooled circular–linear $r = 0.295$), is genome-wide (70.4% of 7,083 genes significant at BH-FDR < 0.05), concentrates in coherent epithelial-polarity and estrogen-signalling modules, and is strongly tumour-specific ($r = 0.295$ in tumours vs. 0.029 in normals). We have also drawn the boundary of the claim honestly: the coupling is symmetric at the cohort level, so these cross-sectional data establish the coupling but not a directional translational flow—the decisive next experiment is a matched multi-omics time course, and the natural extension is a pan-cancer sweep.

The broader point is methodological. Post-transcriptional regulation is exactly the kind of cross-scale, directional relationship that a magnitude-space integration does not natively express, and that a phase-geometric representation does. By encoding each modality as a phasor and reading the coupling with the circular-statistics machinery and its surrogate null, BioPhasor turns a qualitative expectation into a quantitative, reproducible measurement. This is the first use-case study built on the shared BioPhasor platform; the formulation is established once in the companion method paper, and further use cases—a spectral connectome reading of the regulatory axis, port-Hamiltonian dynamics of the phasor state—build on the same code and the same geometry rather than re-deriving them.

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